10×100 $\quad 100 \times 5$ $\quad 5 \times 50$

$\langle A_1, A_2, A_3 \rangle$

$(A)_{p \times q}$ $(B)_{q \times r}$ → pqr scalar multiplications

$(A_1 A_2) A_3 = A_1 (A_2 A_3)$ – Associativity.

$(A_1 A_2)$: 10×100, 100×5 → 10×5

$(A_2 A_3)$: 100×5, 5×50 → 100×50

$A_1 (A_2 A_3)$: 10×100, 100×50

$10 \cdot 100 \cdot 5$

$+ \; 10 \cdot 5 \cdot 50$

7500

scalar multiplications

$100 \cdot 5 \cdot 50$

$+ \; 10 \cdot 100 \cdot 50$

75000

scalar multiplications

Computing perspective → Minimize no. of scalar multiplications

(↑ Require only 10% Compared to ↑)

Matrix chain Order (Multiplication)

Statement: Given a chain $\langle A_1, A_2, \ldots, A_n \rangle$ of n matrices where for $i = 1, 2, \ldots n$ matrix A_i has dimension $p_{i-1} \times p_i$; we need to fully parenthesize the product $A_1 A_2 \ldots A_n$ in a way that minimizes the no. of scalar multiplications.

Recursive Version:

$$m[i,j] = \begin{cases} 0 & \text{if } i = j \\ \min\limits_{1 \le k < j} \{ m[i,k] + m[k+1,j] + p_{i-1} p_k p_j \} & i < j \end{cases}$$

$m[i,j]$: $T(N)$

size $= i + j = N$ $\qquad i + k = N - j + k$

$$T(N) = T(N-j+k) + T(N-i+k+1)$$

$$\geq T(N-j) + T(N-i)$$

$$\geq 2\,T(N-j)$$

↓

Gives exponential order.

Prof. Kedukodi Babushri Srinivas, Mathematics, SBHM, MIT Manipal

Dynamic Programming Soln:

Matrix-Chain-Order(P):

Input sequence $P = \langle p_0, p_1, \ldots, p_n \rangle$

length[P] = n+1.

P_i has dimension $p_{i-1} \times p_i$

1. $n \leftarrow$ length[P] − 1.
2. For $i \leftarrow 1$ to n
3. $m[i, i] \leftarrow 0$.

4. For $l \leftarrow 2$ to n // l is Chain length

5. Do for $i \leftarrow 1$ to $n - l + 1$

6. $j \leftarrow i + l - 1$

7. $m[i, j] \leftarrow \infty$.

8. For $k \leftarrow i$ to $j - 1$

9. $q \leftarrow m[i, k] + m[k+1, j] + p_{i-1} p_k p_j$

10. If $q < m[i, j]$

11. then $m[i, j] \leftarrow q$

12. $s[i,j] \leftarrow k$.

13. return m and s.

Solve the matrix - chain - order problem

given:

Matrix	Dimension	
A_1	10×100	$p_0 \times p_1$
A_2	100×5	$p_1 \times p_2$
A_3	5×50	$p_2 \times p_3$
A_4	50×20	$p_3 \times p_4$